```
public class Vacuum {
   public static void main(String[] args) {
      char A = 'D', B = 'D';
      int pos = 0;
      while (A == 'D' || B == 'D') {
         if (pos == 0) {
            if (A == 'D') {
               A = 'C';
               System.out.println("Suck Room A");
            } else {
               pos = 1;
               System.out.println("Move Right to B");
            }
         } else {
            if (B == 'D') {
               B = 'C';
               System.out.println("Suck Room B");
            } else {
               pos = 0;
               System.out.println("Move Left to A");
            }
         }
      }
      System.out.println("Both rooms are clean!");
   }
}
```

3. Theory

Intelligent Agent:

An intelligent agent is a system that observes its environment through sensors and performs actions through actuators to achieve a goal.

Basic Structure:

Environment → Sensors → Agent → Actuators → Environment

Vacuum Cleaner World:

It consists of two locations, A and B. Each location can be either Clean or Dirty.

The vacuum cleaner performs the following actions:

Condition	**Action**
Current location is Dirty	Suck
At A and Clean	Move Right
At B and Clean	Move Left

Simple Reflex Agent:

A Simple Reflex Agent selects an action based only on the current percept. It uses condition-action rules.

Example:

- If the current location is Dirty → Suck.
- If the current location is Clean and at A → Move Right.
- If the current location is Clean and at B → Move Left.

Rational Behaviour:

A rational agent selects the best action to achieve its goal. In this case, the goal is to clean both rooms with minimum unnecessary movement and time.

4. Algorithm

1. Start.
2. Define the environment with two locations A and B.

3. Set the initial position of the vacuum cleaner.
4. Check the current location and its cleanliness.
5. If the current location is Dirty, perform Suck.
6. If the location is Clean:
    - If at A, move to B.
    - If at B, move to A.
7. Update the environment after every action.
8. Repeat until both locations are clean.
9. Display the actions and final environment.
10. Stop.

Q1. What is an Intelligent Agent?
An intelligent agent observes the environment and performs actions to achieve a goal.

Q2. What is a Simple Reflex Agent?
It is an agent that takes decisions based only on the current condition.

Q3. What is the difference between an agent and an environment?

- Agent: Performs actions.
- Environment: The surroundings in which the agent works.

Q4. What are sensors and actuators?

- Sensors: Used to observe the environment.
- Actuators: Used to perform actions.

Q5. What is PEAS?
PEAS stands for:

- P – Performance Measure
- E – Environment
- A – Actuators
- S – Sensors

Q6. What is the performance measure of a vacuum cleaner agent?
Maximum cleanliness with minimum time, energy and unnecessary movement.

Q7. Why is a vacuum cleaner considered an intelligent agent?
Because it observes the room and takes actions to clean it.

Q8. What is rational behavior?
Taking the best possible action to achieve the desired goal.

Q9. What are the limitations of a Simple Reflex Agent?
It cannot remember previous actions or past conditions.

Q10. Difference between Simple Reflex and Model-Based Reflex Agent?

- Simple Reflex: Uses only current information.
- Model-Based: Uses current information and maintains internal memory.

Q11. Can a Simple Reflex Agent remember previous actions?
No, it does not remember previous actions.

Q12. What happens if both rooms are clean?
The cleaning goal is achieved, and the agent stops.

Q13. Why is if-else used in this implementation?
To check conditions and select the correct action.

Q14. What are the possible percepts in Vacuum Cleaner World?

- Location A or B.
- Current room Clean or Dirty.